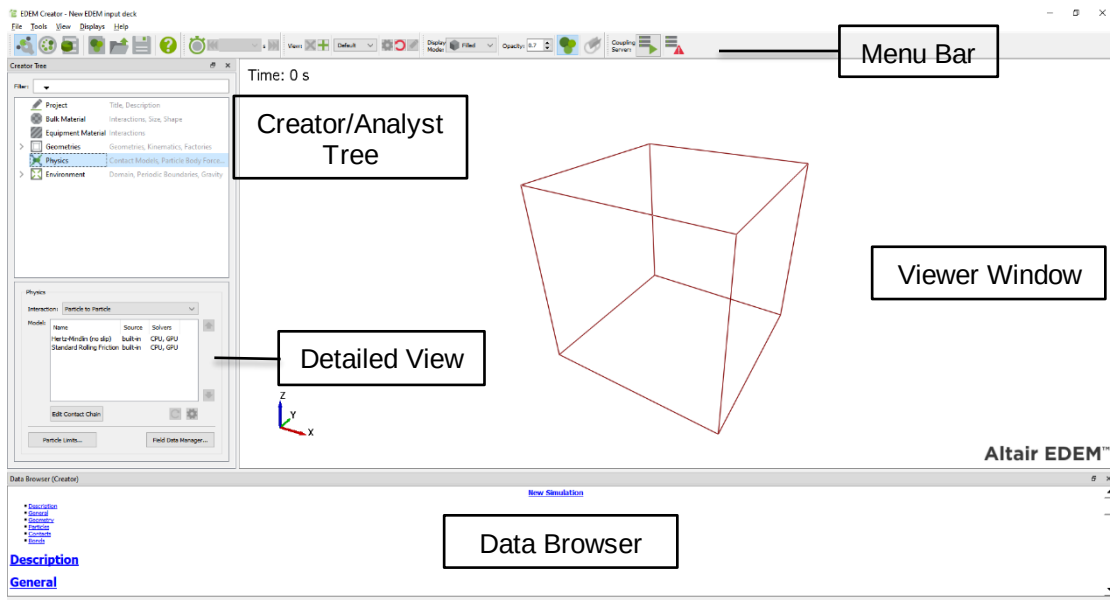


EDEM Application Tutorial

Conveyor

Default EDEM Creator Layout

The EDEM default screen layout is shown below with labels. This layout can be modified by the user if required.



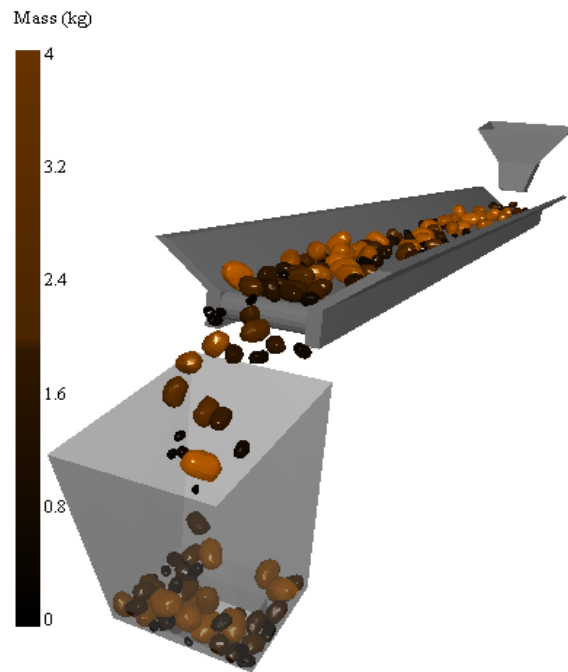
Introduction

Through this introductory tutorial we will learn how to set up and analyze an EDEM simulation. The exercise is directed at new users to get a basic understanding of EDEM.

The model used here is a conveyor belt that transports material to a hopper.

The main focus is on:

- Creation of materials for Particles and Geometry.
- Creation of Particles with a User Defined Size distribution.
- Creation of a MultiSphere Particle Prototype. The multi-sphere particle shape is defined using particle CAD templates, which can be imported to EDEM and manually filled with spherical particles.
- Use Conveyor Kinematics, which provides movement and acceleration of the conveyor belt.
- Creation of bin groups in the Analyst to analyze particles and their properties in specified regions.
- Export of simulation data to a .csv (text) file.



1. Start EDEM.
2. Go to File > Save As...
3. Select a location on the local machine (e.g. **C:\EDEM_Tutorials**).
4. Enter a file name (e.g. **Conveyor_Tutorial.dem**) and click Save.

Please note this tutorial has associated CAD files `conveyor_hopper.stl` and `rock_particle.stl`. It is recommended to copy these files to the same location as the simulation files.

EDEM Creator: Setting up the model

Step 1: Define the Project and Settings

Choose the units

The first step in creating the model is to set the units used throughout EDEM.

1. Go to the Tools > Options... menu and select the Units tab.
2. Change the following measurement units:
 - Angle to degrees (**deg**)
 - Angular acceleration to **deg/s²**
 - Angular velocity to **deg/s**
 - Length to **mm**
 - Mass to **kg**
 - Mass Flow Rate to **kg/s**
3. Click OK.

Enter the model title and description

1. Click on Project in the Creator Tree.
2. In the Detailed View, enter the title **Conveyor Model** in the Title box under the Simulation section.
3. Enter a description in the Description field.

The model title and description will appear in the Data Browser window. The Data Browser window can now be closed if preferred, to gain more space for the windows that will be used throughout this tutorial. If required, open the Data Browser window again by right clicking over the menu bar and ticking the Data Browser from the list.

Step 2: Define the Bulk Material

Next, we add the bulk material which will be used in the simulation.

Add new bulk material

1. Right click on the Bulk Material in the Creator Tree, select Add Bulk Material and enter the name **rock** into the highlighted field. Alternatively, you can also use the icons in the Toolbar.

Define the bulk material properties

1. Select **rock** under the Bulk Material section.
2. Under rock Properties, set Poisson's Ratio, Solids Density and Shear Modulus as shown.

rock Properties	
Poisson's Ratio (v):	0.3
Solids Density (p):	1600 kg/m³
☒ Shear Modulus (G):	1e+07 Pa
☐ Young's Modulus (E):	2.6e+07 Pa

Define the bulk material interactions

1. Click the  button in the Interactions section of the rock Properties. Select **rock** when prompted. This will define the interaction between all elements made of the material rock.
2. Set the coefficients as follows:

Interactions	
Interaction:	rock
Coefficient of Restitution:	0.3
Coefficient of Static Friction:	0.5
Coefficient of Rolling Friction:	0.01

Create a new particle shape

After the bulk material and interaction properties have been defined, the particle shapes for the bulk material need to be created.

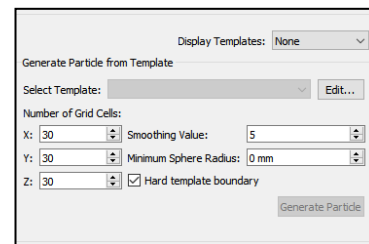
1. Right click on the **rock** material under the Bulk Material section in the Creator Tree, select Add Particle > Add MultiSphere and enter the name **rock particle** into the highlighted field
2. The particle may be renamed by right clicking the particle and selecting Rename Particle.

Define the spheres and properties

Particles are made up of one or more spheres. In this case, we are going to import a CAD template of a rock and fit 4 spheres to the template shape.

1. From **rock particle** go to Select Template > Edit

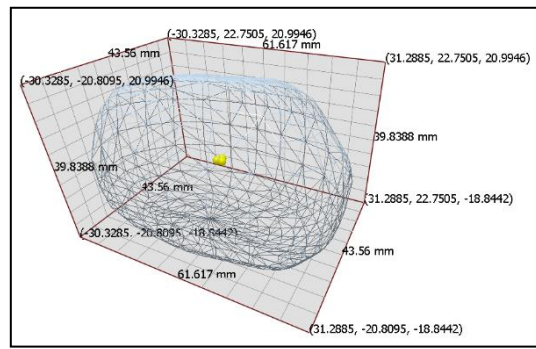
This is found in the bottom right hand corner of the screen.



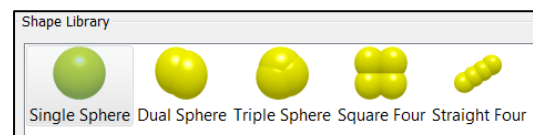
2. Click the Import button then in the Display Templates dialogue box import the particle template **rock_particle.stl**. This is the mesh file we shall use as a particle template.
3. When the Import Options dialog appears, set the measurement units to **Millimeters** (mm). Under Mesh Control type leave all the settings as the default values and click OK.

The STL file format is already meshed, as such changing the meshing options does not impact the mesh of the imported template. For other file formats such as STEP or IGES the geometry/template mesh can be modified by changing the element size options in the import dialog.

4. Click the  button and rename the template to **Rock Template** then click OK.
5. Change Display Templates from the dialog box in the bottom right from Off to Rock Template.



6. Go back to the **rock particle** in the Creator Tree and select Modify Shape in the Detailed View section.
7. Select the Single Sphere from the Shape Library as shown below.



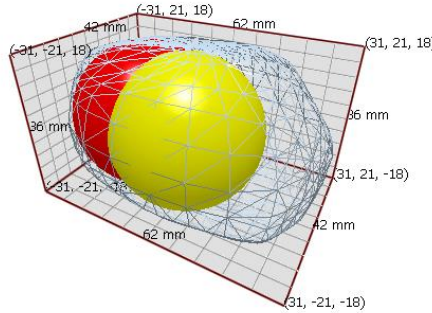
8. In the rock particle Spheres section, under the viewer window, change the Physical Radius of **sphere 0** to **18 mm** and the Position X, Y, Z to **(0, -3, 0)** mm.
9. Add another sphere to the particle shape by right clicking the **rock particle** Spheres table and selecting **Add Sphere**.

rock particle Spheres

☐ Edit Contact Radius

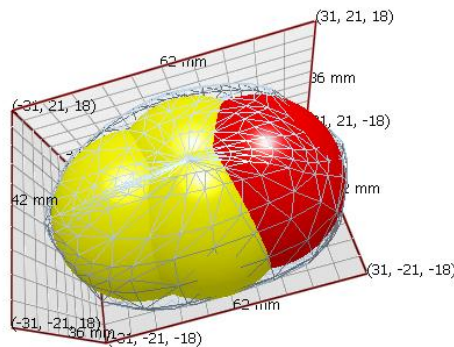
	Name	Position X (mm)	Position Y (mm)	Position Z (mm)	Physical Radius (mm)
1	sphere 0	0	0	0	1

10. Set the Radius of this new sphere (sphere 1) to **18 mm** and the Position X, Y, Z to **(-13, 0, 0)** mm.
11. Having one of the spheres selected from the table in the Rock Particles Spheres section, check the **Highlight Sphere**  box in the View settings in the Menu Bar to be able to distinguish the spheres.

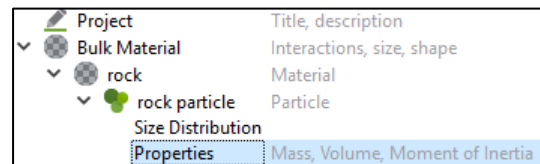


12. Add two more spheres and set their Positions and Radii as below.

	Name	Position X (mm)	Position Y (mm)	Position Z (mm)	Physical Radius (mm)
1	sphere 0	0	-3	0	18
2	sphere 1	-13	0	0	18
3	sphere 2	0	4	0	18
4	sphere 3	12	0	0	18



13. In the rock particle Properties section, tick the **Imported Template** option and click Calculate Properties button.



When using the imported template option, mass, volume and inertia are calculated from the CAD template. Alternatively, you could use the **Spheres** option to calculate these properties from the group of spheres.

Please note that an automated way to generate spheres to fill a template is outlined in Appendix 1. It is recommended in this tutorial to use the manually created 4-sphere particle.

Set the size distribution of the bulk material

1. Expand the **rock particle** section in the Creator Tree and select Size Distribution.

2. In the Size Distribution section, select **user defined** from the drop-down list.
3. Select **Radius** in the Scale By section.
4. Right click in the User Defined Size Distribution Parameters table and select **Add** 3 times. Input the following distribution parameters:

	Scale	% of Mass
1	1	5
2	1.3	15
3	1.6	35
4	1.9	45

Scale By:
☒ Radius
☐ Volume

Total %: **100**

Import

Step 3: Define the Equipment Material

The next step is to add the equipment material which will be used in the simulation.

Add new equipment material

1. Right click on the Equipment Material in the Creator Tree, select Add Equipment Material and enter the name **steel** in the highlighted field.

Define the equipment material properties

3. Select **steel** under the Equipment Material section.
4. In the steel Properties section, set Poisson's Ratio, Solid Density and Shear Modulus as shown.

steel Properties

Poisson's Ratio (v): 0.3

Solids Density (ρ): 7800 kg/m³

☒ Shear Modulus (G): 7e+10 Pa

☐ Young's Modulus (E): 1.82e+11 Pa

Define the bulk material - equipment material interaction

3. Click the button in the Interaction section. Select **rock** when prompted. This will define the interaction between elements made of rock and elements made of steel.
4. Set the coefficients as follows:

Interactions

Interaction: rock

Coefficient of Restitution: 0.2

Coefficient of Static Friction: 0.5

Coefficient of Rolling Friction: 0.01

Step 4: Define the Geometries

The next step is to define the conveyor geometry used in the model.

Import the conveyor geometry

The conveyor geometry has already been created and is ready to be imported into EDEM.

1. Right click on the Geometries section in the Creator Tree.
2. Click on Import Geometry...
3. Navigate to the file **conveyor_hopper.stl** (in EDEM Tutorial 1 – Conveyor.zip) and import it.
4. When prompted, set the Units of measurement to **Millimeters**. Leave the Merge Sections option **unchecked** and leave all the mesh settings as the default values. Click OK.
5. Select then right click or press F2 to rename each section. Name them appropriately:

belt

hopper

lower hopper

guide_1

guide_2

Add conveyor motion

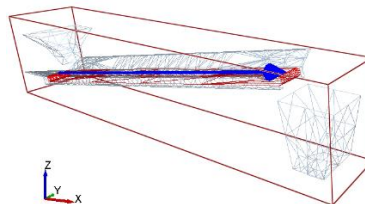
1. Select **belt** in the Creator Tree Geometries section
 - a. Right click > Add Motion > Add Conveyor Translation (New Motion 1) and set up the conveyor as follows:

This creates an acceleration between 1 and 3 s of 1 m/s^2 .

- b. Right Click on the **belt** to add a second Conveyor Translation (New Motion 2) and set up the second motion as follows:

Specification	
Start Time:	3 s
End Time:	9999 s
☐ Loop Every:	0 s
Until Time:	9999 s
Initial Velocity:	2 m/s
Acceleration:	0 m/s ²
Reference Space	
☒ Local	☐ Global
☒ Moves with Body	
Direction	
Start:	Pick End: Pick
X: 0 mm	9848.08 mm
Y: 0 mm	0 mm
Z: 0 mm	1736.48 mm

The second motion is a constant velocity of 2 m/s starting at 3 s. The motion of the belt is set by the Start and End points; in this case the values have been input manually. However, the Pick tool can be used to pick a starting and ending point and the Display Vector tick box shows the direction of the chosen motion.



Create the particle factory plate

Particle factories are used to define where, when and how particles appear in a simulation. All factories must be based on a section of geometry (whether 'real' or 'virtual'). This defines the area or volume in the model that produces the particles.

1. Right click on the Geometries section in the Creator Tree.
2. Select Add Geometry, > **Polygon** and then enter the name **factory_plate** into the highlighted field.
3. Select the polygon and in the General section of the Detailed View set the type to **Virtual**, since the plate is not a physical part of the machinery.
4. Expand the factory_plate section in the Creator Tree and select the Transform subsection
 - Set the position as shown

Transform	
Reference Space	
☒ Parent	☐ Global
Position:	Rotation:
X: 500 mm	0 deg
Y: 0 mm	0 deg
Z: 1500 mm	0 deg

5. Select the Polygon subsection and set the Number of Edges and Dimensions of the polygon as follows:

Rectangle	
Number of Edges:	4
Length Edge A:	800 mm
Length Edge B:	1200 mm

Create the particle factory

1. Right click on the factory_plate section in the Creator Tree.
2. Click Add Factory.

In this model, rock particles are to be created at a rate of 50 kg per second.

1. Select the newly created factory and make sure the Factory Type is set to **dynamic**. If it is not, right click on the new factory in the Creator Tree and click Change Factory Type.
2. In the Particle Generation section, select Total Mass and set the value to **500 kg**.
3. In the Generation Rate section, select Target Mass and set the rate to **50 kg/sec**.

Set the factory's initial parameters

1. In the Parameters section, ensure **rock** is selected from the Material pull-down.
2. Set the Position to **random** to ensure that the particles are randomly created over the entire area of the plate.
3. Set the Velocity to **fixed** and click the configure button  to set the velocity in the z-direction to **-2 m/s**.
4. Select File > Save.

Step 5: Define the Environment

Set the domain of the simulation

1. Select the Environment section in the Creator Tree.
2. Ensure that the Auto Update from Geometry box is checked.

This automatically sets the domain of the simulation depending on the geometries present. It should be set as shown:

Domain	
Min:	Max:
X: -266.081 mm	11515.51 mm
Y: -816 mm	816 mm
Z: -424.4929 mm	2073.782 mm
☒ Auto Update from Geometry	

Set the gravity

1. In the Gravity section of the Detailed View, check that gravity is set to -9.81 m/s^2 in the z-direction.

☒ Gravity
X: 0 m/s ²
Y: 0 m/s ²
Z: -9.81 m/s ²

EDEM Simulator: Running Simulation

Once the model has been set up in the Creator we move to the Simulator to run it.

1. Click on the Simulator button  on the Toolbar.

Step 1: Set the Time options

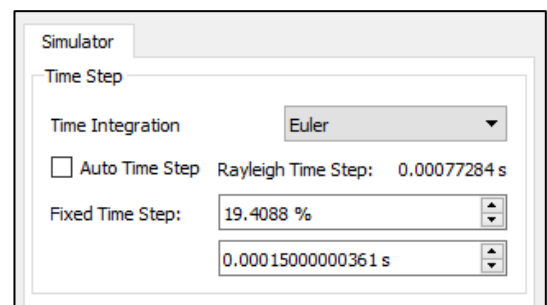
Set the time step

The time step is the amount of time between iterations (calculations) in the Simulator. The time-step value is based on the material properties and is usually set between 20% and 40% of the automatically calculated Raleigh Time-Step. In this simulation a time step of 0.00015 s (~19% of the Raleigh Time Step) should be used.

1. **Uncheck** the Auto Time Step box.

Auto Time Step is a useful feature which automatically adjusts the time step to 20% of the Rayleigh time step throughout the whole simulation. However, for this introductory tutorial, we will use a Fixed Time Step.

2. Set the Fixed Time Step lower box to **0.00015 s**. The percentage of Rayleigh time step will be calculated automatically.

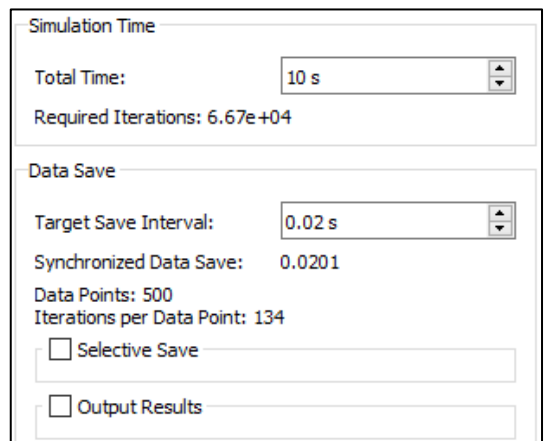


Set the simulation time and data write-out frequency

The simulation time is the amount of real time your simulation represents.

1. In the Simulation Time section, set the Total Time to **10s**.
2. Set the Target Save Interval to **0.02s** to specify the write-out frequency.

It is not usually necessary to write a data point for every iteration in a simulation. Doing so would result in a slow simulation and a very large amount of data being collected.

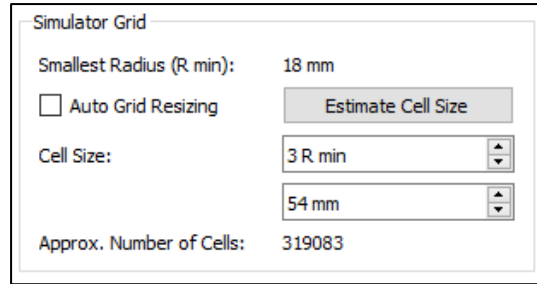


Step 2: Set the Grid options

The grid options are used to optimize the simulation.

1. Set the Cell Size to **3 Rmin**. (Rmin = the smallest particle radius in the simulation)

Note that the number of grid cells (Approx. Number of Cells) is shown. The simulation results are not affected by the number of grid cells.



Simulator Grid

Smallest Radius (R min): 18 mm

☐ Auto Grid Resizing Estimate Cell Size

Cell Size: 3 R min

54 mm

Approx. Number of Cells: 319083

The more grid cells a simulation uses the more system RAM is required. It is necessary to find a balance between RAM usage and simulation performance. For simulations with a narrow size distribution the ideal value will be between 2 – 3 Rmin, but for wider size distributions this will increase, the Estimate Cell Size button will provide an estimate of the optimal grid setting.

Step 3: Run the Simulation

You are now ready to start the simulation.

1. Choose either CPU solver or GPU CUDA Solver from the Selected Engine

As this tutorial has relatively few particles 4 cores CPU solver is appropriate however for larger simulations more cores and GPU CUDA is recommended.

2. Click the start Progress button  at the bottom of the simulation window.
3. In Viewer Controls, click the **Refresh Viewer** button  or enable **Auto Update**  at any point to update the Viewer and see how the simulation is progressing.

EDEM Analyst: Analyzing Your Results

The Analyst is used to review, examine and analyze the results of the simulation.

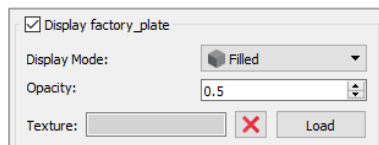
1. Click on the Analyst button  on the Toolbar.
2. Use the controls below the main window    to rewind the simulation to the first time-step.
3. Click the play button  to play through each time step of your simulation.

Step 1: Configuring the Display

The Display section in the Analyst Tree is used to configure how the different elements in your model appear. The geometry, particles and contacts within your model can also be colored in a variety of ways. Any section of geometry, particle, contact type or selection group can be colored independently.

Configure the geometry

1. Click on the Display section in the Analyst Tree and then expand the Geometries subsection.
2. Select different geometries and experiment with altering their display in the Detailed View. For example, try making the hopper transparent, the guides a mesh and the belt fully opaque. Display Mode can be chosen to be **Filled**, **Mesh** or **Points**. The Opacity can take values **0-1**, with 0 being completely transparent and 1 being solid.
3. Before finishing, make sure the **Display factory_plate** box is unchecked, so it is not visible in the Viewer Window.

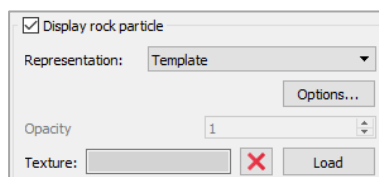


4. Other geometries should be displayed as **Filled** with **0.5** opacity.

Configure the particles

The display of the particles can be altered in a similar way to that of the geometry. In addition, the particles can be represented using vector, cones, stream and particle templates.

1. Expand the Particles section in the Analyst tree and select **rock particle**.
2. Select **Template** from the Representation pull-down menu in the Detailed View.

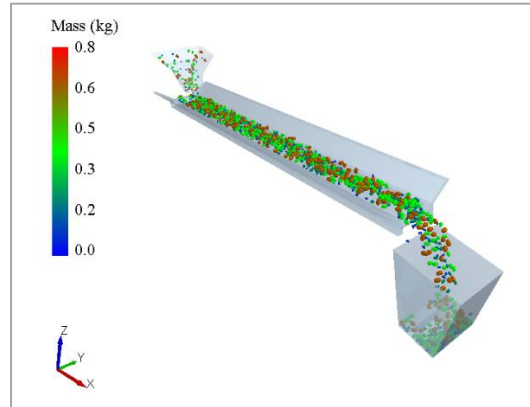


- Click the Options button then set the Template to **Rock Template** and click OK.
In large simulations, a template representation can put a high demand on system resources.
- Watch the simulation step by step by using the controls located at the bottom of the viewer window.
- Set the Representation back to **Default** before proceeding.

Note that the template in the Analyst is for display purposes only and does not change the simulation results. In the Creator the particle mass, volume and inertia can be calculated from a template, if this approach has been taken then the same template should then be used for display purposes.

Color the particles according to their mass

- Select **rock particle** in the Particles subsection of the Analyst Tree.
- In the Coloring section select Color by **Mass** in the pull-down menu.
- Select **3** levels of coloring and set the Min, Mid and Max colors to **Blue, Green** and **Red** respectively. Leave the Smooth Colors button checked.
- Click the buttons  next to the Min and Max Value fields – this reads the current values from the point in the simulation being displayed in the Viewer at the moment. Unselect **Auto Update** and set Min to **0 kg** and Max to **0.8 kg**, this allows for the maximum value in the simulation to be displayed rather than the instantaneous maximum mass value.

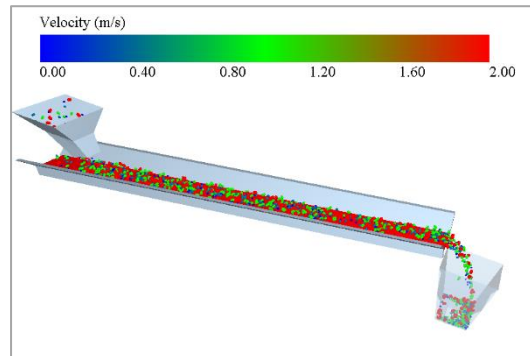


- Check the **Show Legend** checkbox.
- Click the Play button to view the distribution of mass over time.

Color the belt geometry by velocity

- Expand the Geometries subsection in the Analyst Tree.
- Select the **belt** geometry element.
- In the Coloring section of the Detailed View, select **Velocity** in the Color by pull-down menu.
- Set the Min, Mid and Max colors to the **Dark Grey, Green** and **Red** respectively.

5. Set the Min and Max values to 0 m/s and 2 m/s.
6. Play through the simulation to see how the belt velocity changes with time. This data can also be graphed and exported.



Step 2: Creating Selections

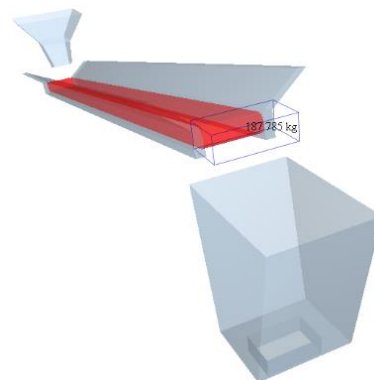
Selections allow data to be extracted from a particular area or element in the domain. Using selections, you can monitor any element passing through a particular area (known as a “bin”) or track particular elements as they move within the domain. You can then display, color, graph or export data based on these selections.

Create a grid bin group

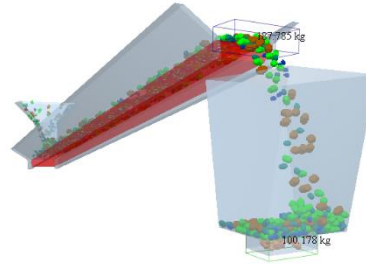
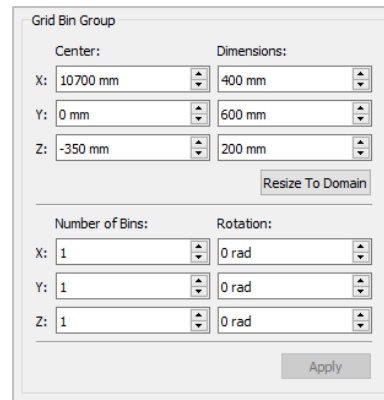
We can compare the total mass of particles that fall off the end of the conveyor belt with the mass of particles falling out the bottom of the lower hopper. To do so, create two grid bin groups each containing one bin.

1. Right click on Setup Selections in the Analyst Tree.
2. Hover on Add Selection > **Grid Bin Group** and name it **Conveyor_group**.
3. Select the newly created grid bin group and in the Detailed View, set its properties as follows then click Apply:

Grid Bin Group	
Center:	Dimensions:
X: 10188 mm	425 mm
Y: 0 mm	800 mm
Z: 1725 mm	250 mm
Resize To Domain	
Number of Bins:	Rotation:
X: 1	0 rad
Y: 1	0 rad
Z: 1	0 rad
Apply	



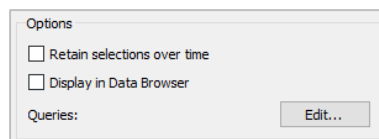
4. Right click on Setup Selections one more time to create a second **Grid Bin Group** and name it **Hopper_group**.
5. Set the properties of the bin group as follows then click Apply:



Define selection group queries

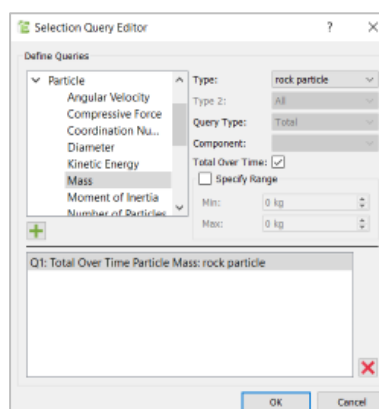
A query is used to define a single element attribute, for example particle velocity, total force on a geometry section or number of collisions. To compare the total mass of particles passing through the first bin group with the number passing through the second:

1. Set the Current Time to about 6.3s. This can be done by using the controls below the viewer window or by typing the current time desired in the time step tool on the menu bar. This is the last time-step before any particles leave the conveyor belt.
2. Select the **Conveyor_group** under Setup Selections in the Analyst Tree.
3. In the Options section of the Detailed View, click the Queries Edit... button:



The Selection Query Editor is displayed.

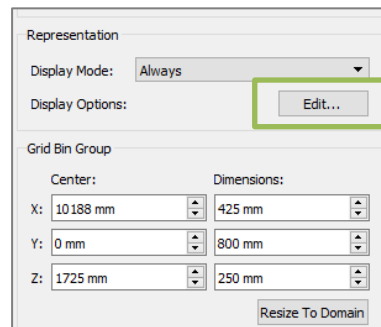
4. Expand the Particle tree, then select **Mass**.
5. Set Type to **rock particle** and Query Type to **Total**.
6. Select the **Total Over Time** option then click the  button and click OK. The query is displayed in the list at the bottom of the dialog:



7. Repeat the above steps to define an identical query for the **Hopper_group**.

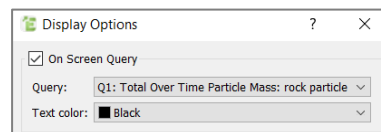
Set selection group display options

- Under Setup Selections in the Analyst Tree Select the **Conveyor_group**, then in the Representation section of the Detailed View click the Display Options Edit button.



The Display Options window is displayed.

- Tick the On-Screen Query checkbox to display the mass in the viewer:



- Repeat the above steps for the **Hopper_group**.
- Click the play button to view the results in real time. As playback continues through the time-steps, bin group data is collected and displayed.

Step 3: Extracting Data

Data collected during your simulation can be exported for analysis. Data can be exported from a single time step or a range of time steps; from all elements in a model or just from a selection or bin group.

Create an export configuration

- Select File > Export > Results Data. The Export Results Data dialog will appear.
- Click the  button under Configuration ID to create a new configuration. A single configuration can contain multiple queries and can be reused at any point in your model.
- Set the Export format to **Text** to extract data to a .csv file.
- Enter the file name **Conveyor** then set the save location to the tutorial folder.

Define a query to export bin group data

Export Total Over Time queries for each bin group.

- In File > Export > Results Data dialog, go to the General tab.
- Tick the **Output Data in Columns** checkbox.
- In the Time Steps section, uncheck **All** then set the range to start at about **6.3s**.

4. Click the Queries tab to open the Queries panel. This is where you define the data to be exported.
5. Click the  button to create a new query.
6. Click on the Particle element in the list then select **Mass**.
7. Set the Type to **rock particle**.
8. Select **Conveyor_group** as the Selection and Query Type to **Total Over Time**.
9. Repeat the instructions 5-8 to add a second query for the **Hopper_group**.

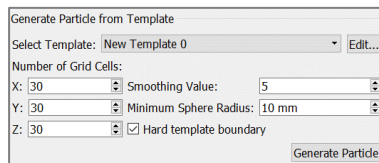
Export the data

1. Click the Export button to export the data to the pre-defined file (Conveyor.csv).
2. Open the file in an external application (such as a spreadsheet) and review the data.

Appendix 1 – Generate Particle from Template

In Step 2: When defining the Bulk Material, the user manually inputs the positions of the particles in the template. An alternative option of “Generate Particle from Template” is available. To use:

1. Select Add Bulk Material from Bulk Material in the Creator
2. Select the new Bulk Material and right click to Add Shape from Library > Single Sphere
3. Under Select Template choose Edit and import the CAD template (if not already imported)
4. Use the following settings for grid cells and sphere radius:



EDEM then automatically generates the particle shape, these settings create approximately 10 spheres to fill the template with the higher fidelity model shown on the left and the manually created 4-sphere template on the right.

